



AI with Rav

Learn AI the way you actually think.



DAY 1 of 30

PYTHON

Why Python for AI + your very first program

WHAT YOU'LL LEARN TODAY

- > What Python is (plain English for computers)
- > Why every AI tool uses it
- > How your code actually runs
- > Your first program, line by line
- > The 30-day journey ahead



Why Python for AI + your very first program

In One Line: Python is like giving instructions to a very obedient helper in plain, simple English. Other languages make you write formal grammar; Python lets you just say what you mean. That's why AI — which needs constant experimenting — was built on it.

Start here: a language that reads like English

Every programming language is a way to tell a computer what to do. But most of them are strict and wordy — full of curly braces, semicolons, and formal grammar. Python is different: it was designed to be **read like plain English**. You write what you mean, and it just works.

This is Day 1 of your Python journey — the first floor of becoming an AI Engineer. We start from absolute zero (no experience needed), and if you already code in another language, I'll give you quick bridges along the way. By Day 30 you'll be ready to start Machine Learning. Let's begin.

Python vs other languages — see the difference

Python = giving instructions in plain, simple English

Other languages (e.g. Java): print a line

```
public class Main {  
    public static void main(String[] a){  
        System.out.println("Hello");  
    }  
}
```

lots of formal grammar

Python: print a line

```
print("Hello")
```

just say what you mean — one line

The same task — printing one line — in Java takes 5 lines of formal grammar (class, main method, System.out.println). In Python it's ONE plain line: print("Hello"). Python lets you just say what you mean.

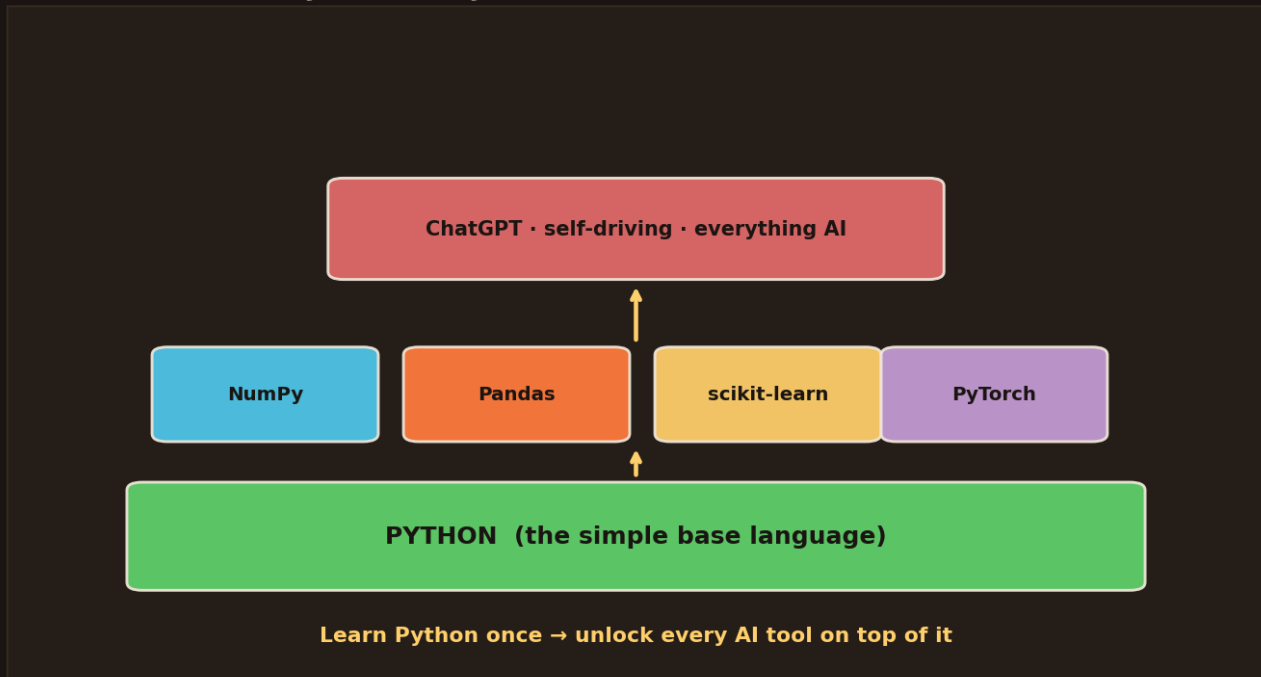
- In many languages (like **Java**), printing one line needs a whole wrapper — a class, a `main` method, `System.out.println`.
- In **Python**, the same thing is just: `print("Hello")` — one line.

(If you already code: yes — no class, no `main`, no semicolons, no type declarations to start. Python cuts the ceremony so you focus on the idea.) **For a beginner, this means you can write real, working code on Day 1. That simplicity is Python's superpower.**



Why every AI tool is built on Python

Why AI loves Python: the whole AI world is built on it



Python is the simple base language. On top of it sit NumPy, Pandas, scikit-learn, PyTorch — and on top of THOSE sit ChatGPT, self-driving cars, everything AI. Learn Python once, and you unlock the whole stack.

Why Python for AI specifically? Because the **entire AI world is built on top of it**:

- The number-crunching tools (**NumPy, Pandas**) are Python.
- The machine-learning tools (**scikit-learn, PyTorch, TensorFlow**) are Python.
- So the big things built on those — **ChatGPT, self-driving cars, recommendation engines** — all start with Python.

This is the key insight: **learn Python once, and you unlock every AI tool on top of it.** Researchers chose Python because it's fast to experiment in — and once everyone used it, all the tools were made for it. It's now the undisputed language of AI. Learning it is the smartest first move you can make.

How your code actually runs



How your code runs



Python reads your plain-English instructions and does them, top to bottom

Three steps: YOU WRITE the code (e.g. `print("Hi")`) → PYTHON reads and runs it (the "interpreter") → you get the RESULT (Hi). Python does your instructions top to bottom.

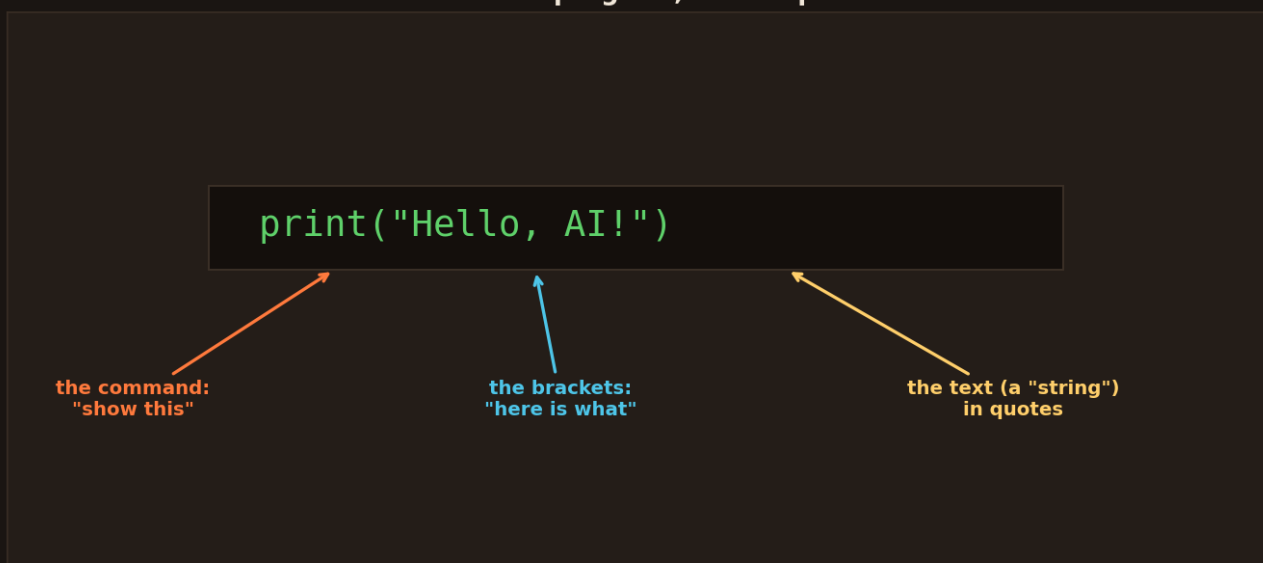
Before we write code, understand what happens when you "run" it:

- **You write** instructions in a file (or a notebook).
- **Python reads them and runs them** — the thing that does this is called the **interpreter**.
- You get a **result** — printed text, a calculation, whatever you asked for.

Python reads your instructions **top to bottom**, one line at a time, and does exactly what each says. There's no hidden magic. If you can write a clear list of steps, you can write Python. That's the whole mental model.

Your first program, line by line

Your first program, taken apart



Breaking down `print("Hello, AI!")`: `print` is the command ("show this"), the brackets `( )` hold what to show, and `"Hello, AI!"` is the text (a "string") inside quotes.



Here it is — your first real Python program:

```
print("Hello, AI!")
```

Let's take it apart, piece by piece:

- **print** → the command that means "show this on the screen."
- **()** → the brackets that hold *what* to show.
- **"Hello, AI!"** → the text you want shown. Text wrapped in quotes is called a **string** (we'll go deep on strings on Day 3).

That's it — you just wrote a program. Run it and the screen shows: `Hello, AI!` Change the words inside the quotes and it shows those instead. Try `print("I am learning Python")`. This tiny command is something you'll use every single day to see what your code is doing.

A few more first steps

```
print("Line one")
print("Line two")           # each print goes on its own line

print(2 + 2)                # you can print numbers and maths too -> 4

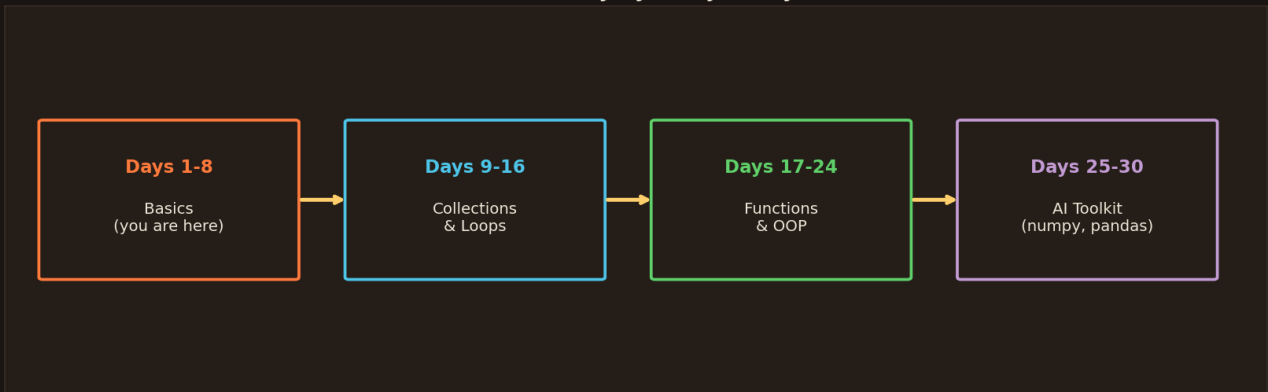
# a line starting with # is a COMMENT – Python ignores it.
# comments are notes for humans, not instructions.
```

Two things to notice: (1) each `print` makes its own new line, and (2) anything after a `#` is a **comment** — a note Python ignores, written for humans reading the code. Good code has comments explaining the "why." You'll thank yourself later.

Your 30-day journey



Your 30-day Python journey



The 30-day Python path: Days 1-8 the Basics (you are here), Days 9-16 Collections & Loops, Days 17-24 Functions & OOP, Days 25-30 the AI Toolkit (numpy, pandas). Each builds on the last.

- **Days 1-8** → the basics (you're here): variables, text, numbers, decisions.
- **Days 9-16** → collections (lists, dictionaries) and loops.
- **Days 17-24** → functions and object-oriented programming.
- **Days 25-30** → the AI toolkit: NumPy, Pandas, plotting — ready for Machine Learning.

We take it **one small step per day**. By the end you won't just "know Python" — you'll be able to load data, clean it, and analyze it like an AI engineer. Everything else in AI is built on this foundation, so building it well is the best investment you'll make.

Recap — the 20-second version

- Python is **plain-English instructions** for a computer — the simplest language to start with.
- **All of AI is built on Python** — learn it once, unlock every AI tool.
- You write code → the **interpreter** runs it top to bottom → you get a result.
- **print("...")** shows text on screen; **#** starts a comment Python ignores.
- This is Day 1 of a 30-day journey that ends with you ready for Machine Learning.

Next up — Video 2: Variables & Data Types. Now that you can print things, let's learn how to STORE things — names, numbers, yes/no values — in labelled boxes called variables. The building block of every program. See you tomorrow.

